

Java Exception Handling Lab Report

Program: IOException

1. Aim

To demonstrate handling of IOException in Java while reading the first line of a file using BufferedReader and try-with-resources.

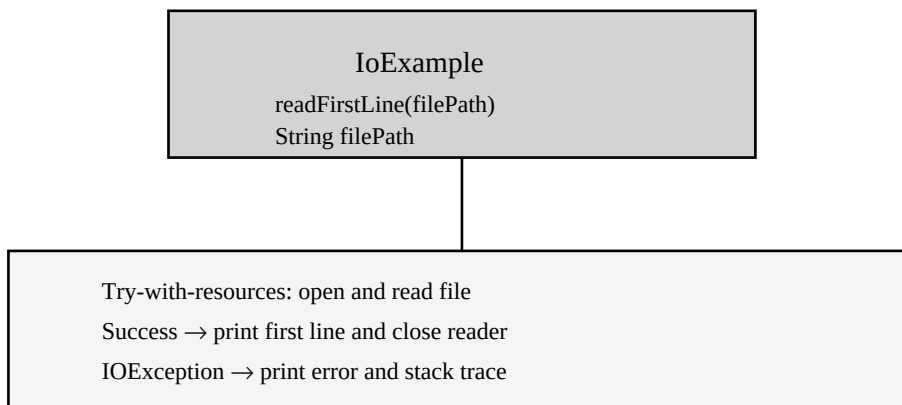
2. Description

IOException is a checked exception that can occur during input and output operations, such as reading a missing, unreadable, or inaccessible file. This program uses FileReader and BufferedReader to read the first line of a file. The try-with-resources statement automatically closes the reader after use. If an I/O error occurs, the catch block prints an error message and the exception stack trace.

3. Tools Used

- Java Development Kit (JDK)
- Java compiler (javac)
- IDE or command prompt
- BufferedReader, FileReader, and IOException from java.io package

4. UML / Program Structure



5. Algorithm

1. Start the program.
2. Define the `readFirstLine()` method with a file path parameter.
3. Create a `FileReader` and `BufferedReader` using `try-with-resources`.
4. Read the first line of the file using `readLine()`.
5. Display the first line.
6. Automatically close the reader when the try block finishes.
7. If an input/output error occurs, catch `IOException`.
8. Display the file path, print the error stack trace, and stop.

6. Complete Java Source Code

```

package exceptions;
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.IOException;

public class IoExample {
    public void readFirstLine(String filePath) {
        // Try-with-resources automatically closes the BufferedReader
        try (BufferedReader reader = new BufferedReader(new FileReader(filePath))) {
            String line = reader.readLine();
            System.out.println("First line: " + line);
        } catch (IOException e) {
            // Handle external failure (missing file, unreadable drive, network drop)
            System.err.println("Failed to read file at path: " + filePath);
            e.printStackTrace();
        }
    }
}

```

7. Sample Output

Successful file read (example):

```
First line: Welcome to Java
```

If the file cannot be read (example):

```
Failed to read file at path: abc.txt
java.io.FileNotFoundException: abc.txt (The system cannot find the file specified)
```

Note: The exact exception message and stack trace depend on the file path and operating system.

8. Result

The program demonstrated `IOException` handling while reading a file. It used `try-with-resources` to close the reader automatically and handled file-reading failures through the `catch` block.